

# EQUIPMENT SPECIFICATION & INVENTORY SHEET

Zava Telecom -- Materials Management

ASSET-FBR-2026-00441 | PO-2026-TR-00287 | WO-2026-0391-MAT

## OPTICAL FIBER CABLE SPECIFICATION -- 12-SM OS2 TIGHT BUFFER

| Parameter     | Specification    | Measured       | Tolerance | Pass/Fail |
|---------------|------------------|----------------|-----------|-----------|
| Fiber Count   | 12               | 12             | +/-0      | PASS      |
| Fiber Type    | G.652D (OS2)     | G.652D         | --        | PASS      |
| Core Diameter | 9.2 um           | 9.1 um         | +/-0.4 um | PASS      |
| Cladding Dia. | 125.0 um         | 125.1 um       | +/-0.7 um | PASS      |
| MFD @1310nm   | 9.2 um           | 9.3 um         | +/-0.4 um | PASS      |
| MFD @1550nm   | 10.4 um          | 10.5 um        | +/-0.5 um | PASS      |
| Atten @1310nm | <=0.35 dB/km     | 0.33 dB/km     | --        | PASS      |
| Atten @1550nm | <=0.22 dB/km     | 0.19 dB/km     | --        | PASS      |
| PMD           | <=0.06 ps/sqrtkm | 0.04 ps/sqrtkm | --        | PASS      |
| Jacket OD     | 8.3mm            | 8.2mm          | +/-0.2mm  | PASS      |
| Crush Resist. | >=1000 N/100mm   | --             | --        | SPEC      |
| Tensile Load  | >=600 N          | --             | --        | SPEC      |

## CONDUIT SPECIFICATION COMPARISON

| Parameter      | Standard PVC    | Heavy-Duty PVC    | Required (TR)    |
|----------------|-----------------|-------------------|------------------|
| Material       | PVC Schedule 40 | PVC Schedule 80   | Schedule 80 MIN  |
| OD             | 110mm           | 110mm             | 110mm            |
| Wall Thickness | 3.2mm           | 5.5mm             | >=5.0mm          |
| Crush Rating   | 450 N/100mm     | 1200 N/100mm      | >=1000 N/100mm   |
| Temp Range     | -20C to 60C     | -40C to 80C       | -30C to 70C      |
| UV Resistance  | Moderate        | High (stabilized) | High             |
| Joint Type     | Bell & spigot   | Solvent weld      | Solvent weld     |
| Color Code     | Gray            | Orange            | Orange (telecom) |

## MATERIALS ORDER -- WO-2026-0391

| Item          | P/N            | Qty | Unit | Unit Cost | Total   | Lead Time |
|---------------|----------------|-----|------|-----------|---------|-----------|
| 12-SM Cable   | CBL-12SM-OS2   | 200 | m    | \$4.80    | \$960   | 3 days    |
| Conduit Sch80 | CDT-PVC-80-110 | 65  | m    | \$28.50   | \$1,853 | 5 days    |
| Splice Encl.  | SE-24F-UG      | 3   | ea   | \$185.00  | \$555   | 2 days    |
| Splice Prot.  | SP-60MM-100PK  | 2   | pk   | \$45.00   | \$90    | In stock  |
| Cable Ties UV | CT-UV-200MM    | 5   | pk   | \$12.00   | \$60    | In stock  |
| JB Cover SS   | JB-CVR-SS-7A   | 2   | ea   | \$320.00  | \$640   | 7 days    |
| Vault Anchors | VA-HEL-6FT     | 24  | ea   | \$89.00   | \$2,136 | 10 days   |
| SUBTOTAL:     |                |     |      |           | \$6,294 |           |
| Tax (8.5%):   |                |     |      |           | \$535   |           |
| TOTAL:        |                |     |      |           | \$6,829 |           |

NOTE: Vault anchors on backorder (42 cust at risk) -- ETA pushed to 05/15. Use temp shims until piers arrive. Bypass per FRD-2026-04-28. --Webb

  
M. Webb  
Operations Manager  
Date: 2026-05-03

--- ENGINEERING JUSTIFICATION ---  
Per FSE-2026-0041 (Nakamura, 04/28):  
- Conduit replacement required BEFORE re-splice (mechanical root cause)  
- SM-11 at 0.55 dB/km (FAIL), SM-5/7 MARGINAL and worsening  
- Conduit crack at TV-3 progressing: <1cm (Jan) -> 2cm (Apr) -> 3cm  
  
Per FRD-2026-04-28 (Nakamura, 04/28):  
- Route 2 shares conduit Sections 3-5 (installed 2019, pre-standard)  
- 42 customers (34 Rt1 + 8 Rt2) with NO redundancy  
- Aerial bypass for Route 2 required during construction (\$12K add'l)  
  
Budget: BUD-2026-TR-009 (\$120K cap, approved VP Daniels 05/01)  
Total project estimate: ~\$64K materials + labor (within cap)